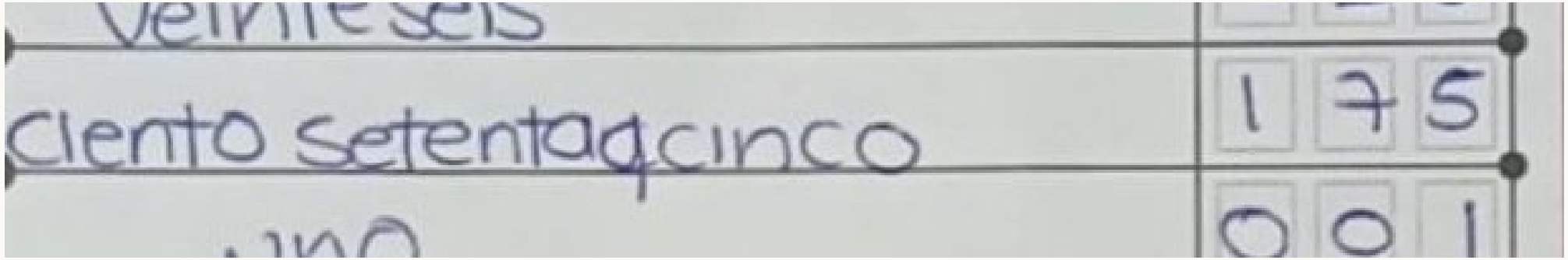
Two Hands: receipts for every number keyed from paper

Two hands: two independent readers, and the paper's own arithmetic as the referee.

A second keyer that shows its proof. An open vision model, **Gemma on AMD**, re-reads every field of forms already keyed; the paper's arithmetic decides what is accepted; a person sees every disputed field with its photo.

Measured on a pre-registered random 300 of Mexico's 2024 handwritten tally sheets. An independent demonstration of clerical capture quality, not an official INE product.

One receipt



THE PAPER

175

words "ciento setenta y cinco" = one hundred seventy-five · boxes 1 7 5

THE FIRST CAPTURE KEYED

~~75~~

one of 52 fields where the capture does not match the paper

THE CERTIFIED COUNT

175

settled by comparison with the paper ("Cotejo")

Form 050815B0100, Option 07, from the pre-registered random 300 (masked crop). Sources: app/data/headline.json ("receipt"); INE PREP 2024 and Cómputos Distritales 2024, free-use terms. Each receipt is Merkle-anchored: "verify this receipt" in the app.

The problem, from INE's own files

6.5%

of the 52,167 polling stations that were not recounted: INE's two official captures of the same paper disagree on at least one vote field (3,382).

19.2%

of fully captured national forms fail their own row sum in the official capture (26,336 of 137,196).

software, not values

The mandated outside audit of each PREP, by rule, checks the software, not the keyed values against the paper.

A difference says two captures disagree, not which one is right; the photo does.

The preliminary count is preliminary by design; the certified count is the official result.

Sources: results/national_counts.md (PREP 2024 CSV; Cómputos Distritales 2024 CSV); audit scope: Reglamento de Elecciones art. 347.1 and INE's 2024 PREP audit report (UAM-I), PDF p. 7. Clerical only: no party, no totals, no outcome.

How it works: two hands and a referee

HAND 1

Gemma-4-31B-IT on AMD reads every field twice, the handwritten digits and the amount in words, plus each row's printed label.

HAND 2

A second, independent reader (Qwen3.8-27B on AMD) must agree field by field. It never overrides Gemma; it only withholds acceptance.

THE REFEREE

The paper's arithmetic: words = digits, rows sum to TOTAL, sections agree, every row label where it belongs.

The gate. Every identity holds: accepted. Any fails: the form is routed, with the failing field named. It never guesses a value.

A person sees the rest: the field, both reads and the crop. One value typed; the identities re-check at once. Names column masked before anything is stored or sent.

A form type is one schema file. Pipeline: locate the page and mask it (CPU) → one request per form on AMD Token Factory → gate (plain Python). docs/ARCHITECTURE.md.

Measured, not promised

52
fields where the official capture does not match the paper, each with its photo; **22** confirmed by the certified count (of the 31 it covers)

ONE READER (PRIMARY)

18%
of forms accepted with no person: 2 wrong of 1,210 fields

TWO READERS

30%
0 wrong of 1,958 fields

CONFIRMATION, 100 FRESH FORMS

0 / 484
accepted fields wrong (Wilson 95% upper bound 0.79%); pre-registered bar ≤ 0.5% met on the point estimate

Alone, one model mis-keys 1 field in 37. Behind the gate, 1 in 605 accepted fields is wrong.

Pre-registered random 300 (protocol a93294c, Amendments 1-3, committed before any model read); Gemma-4-31B-IT read 289. The two-reader tier checks words = digits and row sums, without the section checks. Confirmation: gate v2, seed 20261017, results/confirm100.md (truth: AI adjudicator on published crops, labelled). app/data/headline.json; results/two_reader_300.md.

The buyer already budgeted for this

- INE put MXN 9.5M in its 2024 draft budget for an automated capture service. (p. 4)
- IPN quoted MXN 19.9M for the service. (p. 4)
- The acceptance bar was 80% agreement on words and digits. (p. 5)
- The best entrant reached 75%; INE declined it as untested in a real election, and its advisers (COTAPREP) recommended a pilot. (pp. 5, 10)

The pilot INE's advisers recommended can run on 2024's own forms today.

83.5%

On that metric as we read it, Gemma's existing reads of our pre-registered 300 agree with the official capture in words and digits on 83.5% of fields (5,508 of 6,600; failed reads and blank or illegible captures count against us).

INE's 2024 acceptance metric as we read it from the report, computed on our existing reads after they were made; the definition was committed before computation; not the same test set or conditions as INE's trial.

Beside it, the result that decides acceptance: 0 wrong of 1,958 accepted fields (two readers, the paper's sums as referee).

Source: INE, PREP PEF 2023-2024, "Captura Automática" report, CCOE 10th session 26-08-2024 (repositoriodocumental.ine.mx, ccoe-10se-26-08-2024-p17-1.pdf). Metric: results/ine_metric_DEFINITION.md (committed before computation), results/ine_metric.md. Buyer: the state institute (OPLE); PREP vendors and outside auditors are the channel (docs/BUSINESS.md).

Price and unit economics

LINE 1 (PROPOSAL)

$1

$1 per form audited (second capture + cotejo + receipts)

LINE 2

$5 per confirmed capture error, $0 if none is found

Costs per form. AMD compute $0.00014 per form (measured) · review labour for routed fields about $0.09 to $0.13 per form (GUESS: 20 s per disputed crop) · hosting and sales, not measured.

Anchor. INE's own line was MXN 9.5M for ~170,000 actas, about MXN 56 (~$3) per acta for a first capture alone.

Per state.

- Morelos, 7,802 forms: about $7,800 (MXN 143k), under the lowest federal direct-award ceiling
- the largest state (Estado de México), 20,908 forms: about $20,900 (MXN 384k), over the lowest federal direct-award ceiling, under the highest
- the federal count, 169,005 forms: about $169,000 (MXN 3.1M), over every federal direct-award ceiling: a tender; INE's own line was MXN 9.5M

Line 2 basis: 52 confirmed capture errors in 267 disputed fields on 289 read forms, $1 / 0.180 = $5.56, rounded down. Review: 0.92 crops per form measured, $18 to $25 an hour. FX: MXN 18.3835 per USD (ECB, 2026-10-16). Ceilings: PEF 2026 Anexo 9. One config: app/pricing.py; docs/BUSINESS.md.

Gemma on AMD

Gemma-4-31B-IT is the primary reader in every mode. Every form of the 300, on AMD GPUs via AMD Token Factory: one request per form (page + 6 zoomed tiles).

AMD COMPUTE PER FORM
$0.000144

MEAN LATENCY PER FORM
87 s
shared endpoint, ~2 forms/min

TOKENS PER FORM
2,609 / 804
in / out; 289 of 289 replies end in stop

Open-model bake-off, same forms, prompt and gate: Gemma-4-31B-IT is the per-dollar leader and accepts the most forms. Gemma + Qwen3.8-27B is the safest pair: 0 wrong in 1,958.

Open model on AMD Token Factory	Gemma-4-31B-IT	Qwen3.8-27B	Qwen3.8-Flash-Next
Forms accepted, no person	**55 of 300**	31 of 300	17 of 128
Cost per 1,000 forms	**$0.14**	$0.64	$2.11
Mean latency per form	**87 s**	241 s	120 s

results/bakeoff_300.md (Amendment 6), results/gemma_vs_qwen.md, results/calls_gemma.jsonl; docs/GEMMA.md. Gemma was withdrawn from Token Factory 2026-10-15 15:41 UTC and back 2026-10-16 02:06 UTC (endpoint log); every measured read of the 300 is Gemma on AMD.

Evolus: where the case lives

Workspace py_tr (event code AMDACT3).

- **Agent "Intake clerk"** on Gemma-4-31B-IT, AMD Token Factory: drafts each finding "for the auditor to confirm". It never decides.
- **Workflows:** "Audit finding" (the live audit opens a case per strong-evidence dispute, at most 60 a day), "Review hand-off", "Record filing".
- **AI Tool** `translate`: every review task in English and Spanish. **Scripts** re-check the paper's identities with no AI.

REQUIREMENT 2, CONFIRMED

"Token Factory works fine on Evolus and does count for the Track."

The Evolus team, Discord, 2026-10-16

The decision stays with the paper's arithmetic, never with a model; Evolus records each run as a workflow audit.

docs/EVOLUS.md; evolus/ (built and driven over the Evolus REST API, docs/evolus-openapi.json). Workspace and model endpoint stay online until Nov 15, 2026 for the Evolus review.

The next markets, not measured

1. **Health facility reports (DHIS2):** WHO's Data Quality Review recounts paper source documents against what was keyed; DHIS2 runs in 80+ countries; monthly forms carry their own totals.
2. **US mortgage post-closing QC:** lenders re-check at least 10% of closed loans against source documents every month; 4.4 million closed-end originations in 2023.
3. **Clinical-trial source data verification:** it corrects 1.1% of site data while monitoring takes 15-30% of trial costs.
4. **Peru:** ONPE's 2026 rule keys every acta twice, one of them by a virtual assistant; not measured by us.

The referee is arithmetic, and arithmetic has no language. Measured on Spanish handwriting and English ledgers; the only language-specific check is words against digits, and the reader is a multilingual model.

Sources: WHO DQR Module 3 (2020); dhis2.org/about; Kenya MOH 711A; Fannie Mae Selling Guide D1-3-01, D1-3-02; CFPB HMDA 2023; Lindblad et al., Clinical Trials 2014; ONPE RJ 000093-2026 (El Peruano, 6 Jun 2026). Links: docs/BUSINESS.md.

Limits and the failure case

- **SOBRANTES (unused ballots)** is the one graded field in no identity printed on the form: every accepted-field error on the 300 was SOBRANTES. Gate v3 (pre-registered) accepts it only when two independent reads agree; otherwise a person sees it.
- **Digit insertion:** on 070212B0100 the model read 3114 where the paper says 314; the frozen gate accepted it, gate v2 did not.
- **Harmonised reads:** in one call the model can make words and digits agree (010469C0400: words and digits 239, paper 234). One read is not two independent reads; the fix is a second reader.
- **Label-gate false alarms:** the row-label check routed 86 forms; 68 of them were read right on every field.
- **Failure case:** a rainfall ledger read as one image shifted a row one year; sums still held, 26 of 870 accepted cells wrong. Tiles and the row-label check removed it.

Safeguard: accept only when every identity holds; everything else goes to a person with the field named. Fail closed when the page cannot be located (11 of 300).

app/data/headline.json ("harmonised", "label_gate"); results/confirm100.md; results/bakeoff_300.md ("Wrong accepted fields"); README "Limits and failure cases"; docs/EVALUATION.md.

Receipts for every number keyed from paper.

A second keyer that shows its proof: two independent readers on AMD, and the paper's own arithmetic as the referee. Every disputed field goes to a person with its photo.

Live demo: add at submission · **Video:** add at submission · **Repo:** add at submission

Reproduce every number offline: `make eval`. Gemma-4-31B-IT on AMD Token Factory · Evolus workspace py_tr. Vote rows shown as Option 01 to 15; no party is named or compared.